University Group Diabetes Program

A Study of the Effects of Hypoglycemic Agents on Vascular Complications in Patients with Adult-onset Diabetes

VI. Supplementary Report on Nonfatal Events in Patients Treated with Tolbutamide

Careful review of data available on patients enrolled in the University Group Diabetes Program (UGDP) led the study investigators to conclude in 1969 that the combination of diet and tolbutamide therapy was no more effective than diet alone in prolonging life and possibly less effective than diet alone or than diet and insulin insofar as cardiovascular mortality was concerned. The bases for these conclusions were presented in previous reports from the UGDP.[1-3]

Only limited information on nonfatal events for patients treated with tolbutamide and diet compared with patients treated with diet alone and patients treated with diet plus insulin was presented in the earlier reports[2,3] from the UGDP. The purpose of this report is to summarize for these patient groups the data on nonfatal events collected at scheduled follow-up examinations through the follow-up examination at which tolbutamide treatment was discontinued. Particular attention has been given to the occurrence of events indicative or suggestive of cardiovascular disease in view of the excess cardiovascular mortality of approximately 1 per cent per year observed for patients treated with tolbutamide compared with patients treated with diet alone or treated with diet plus insulin.

Prepared for the University Group Diabetes Program by Max Miller, M.D., Genell L. Knatterud, Ph.D., Barbara S. Hawkins, M.S., and William B. Newberry, Jr., M.D.

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METHODS

Design

The design of this prospective study of patients with adult-onset non-insulin-dependent diabetes has been described in detail in earlier reports,[1-3] but a few points will be reiterated here. Results for only the 823 patients randomly allocated to the four treatment groups outlined in table 1 are included in this report. The number of patients in each of the four treatment groups was approximately equal—205 in PLBO (placebo), 204 in TOLB (tolbutamide), 210 in ISTD (insulin standard), and 204 in IVAR (insulin variable). The results for patients allocated to phenformin treatment are not considered in the evaluation of tolbutamide therapy because this treatment was added 18 months after recruitment was started for the other four treatment groups and because it was not used in all clinics. An evaluation of phenformin therapy based on findings for fatal and nonfatal events was presented and discussed in other UGDP reports.[4,5]

Closing Date of Analysis

To include all data on nonfatal events for patients assigned to tolbutamide or to the corresponding placebo treatment group through the quarterly follow-up examination at which the use of tolbutamide was discontinued, the results for all the follow-up examinations performed in the UGDP clinics on or before October 31, 1969, were utilized for all the analyses presented in this report. Results previ-

TABLE 1

Study treatments

Insulin variable (IVAR)	— amount required to maintain "normal" glucose levels.
Insulin standard (ISTD)	— 10, 12, 14, 16 units, depending on patient's body surface.
Tolbutamide (TOLB)	— 1.5 gm. (split dosage).
Placebo (PLBO)	— dosage schedule similar to that used for tolbutamide

ously reported[1-3] for these patients were based on information *processed* by the Coordinating Center as of October 7, 1969.* Inclusion of the data from all examinations *performed* before October 31, 1969, as compared with the information *processed* by the Coordinating Center as of October 7, 1969, adds at least one, and in some cases two, quarterly follow-up examinations for all active patients because some time was required to complete each examination within the clinic and to forward the forms to the Coordinating Center. Additional time was required to obtain the lipid results and the readings of ECGs, fundus photographs, and x rays.

The difference in cut-off dates has a small but insignificant effect on the mortality findings. No deaths occurred among the placebo and tolbutamide-treated patients in the interval October 7 through 31, 1969. However, seven patients, two in the placebo-treated group, three in the insulin standard-treated group, and two in the insulin variable-treated group, died prior to October 7, 1969, but the clinics did not learn of these deaths until 1970. The deaths in each of the insulin groups were attributed to cardiovascular causes. One death in the placebo-treated group was attributed to cardiovascular causes and one to cancer. Autopsy information† on one death in the placebo-treated group that was counted as a cardiovascular death in previous reports[2,3] led to the determination of the review team that this death should be ascribed to noncardiovascular causes.

Data Collected at Each Examination

During the month before initiation of treatment each patient was instructed in an appropriate diet and then given a general physical examination and detailed examinations of the eyes, heart, kidney function, and peripheral vascular system. These data provided the baseline information. At the end of that one-month period, if the patient was free of diabetic symptoms, particularly ketoacidosis, the patient was randomly assigned to treatment and enrolled in the study.

Patients were asked to return every three months for follow-up examinations. At each quarterly follow-up examination, a general clinical examination and one of the detailed examinations for eyes, heart, kidney function, and peripheral vascular system were performed. The eye examination was performed in conjunction with the first quarterly examination of each year of follow-up. The heart examination was coupled with the second quarterly examination, the kidney function tests with the third, and the examination of the peripheral vascular system with the fourth. At each quarterly visit the patient was questioned concerning level of adherence to diet and the use of the assigned medication. The patient was also queried to determine all drugs prescribed and taken since the last completed quarterly follow-up examination and the reason each drug was prescribed. Each drug prescribed and taken was recorded at each quarterly examination and was coded according to a classification system[5] modeled after the one outlined in A.M.A. Drug Evaluations.[6] The reason each drug was prescribed was coded according to the system detailed in a previous report.[5] Patients were queried to determine whether they had been hospitalized since the last completed follow-up examination; the duration in days and the reason for the hospitalization were recorded for each occurrence. Illnesses not requiring hospitalization were also recorded.

Analytic Procedures

The identification of a specified nonfatal event or condition that occurred during the course of the study was based on observations made at scheduled follow-up examinations. The only patients considered at risk of developing a specific condition or having a nonfatal event during the course of the study were those patients who met the following conditions:

(1) The patient was examined for or questioned about the condition (event) during the baseline examination and did not have the condition at that time and/or did not report a history of the event,

and

(2) The patient completed one or more follow-up examinations during which he/she was

*Two deaths (one in ISTD and one in IVAR) after a quarterly examination was performed during the same time interval as the discontinuation examination of tolbutamide-treated patients and prior to October 31, 1969.

†The autopsy report was received in the Coordinating Center on January 20, 1972, and immediately submitted to the review team.

examined or questioned about the condition or event.

For example, 192 of the 205 patients in the placebo-treated group reported no prior history of angina pectoris at the baseline heart examination, but three of these patients did not complete any of the follow-up heart examinations. Thus, only 189 patients in the placebo treatment group were considered at risk of having angina pectoris for the first time during the course of the study and could have reported the occurrence of this event.

In all the analyses presented in this report only the first occurrence of the specified event was considered, so a patient was counted only once for each event. Review of the findings for the first occurrence of an event provided a means of screening for differences among the treatment groups. Whenever treatment differences were noted, additional analyses were performed to determine the persistence and the clinical importance of these findings.

The drug-placebo comparisons (TOLB versus PLBO, ISTD versus PLBO, and IVAR versus PLBO) of the number of patients who had a specified event or were identified with a specific condition for the first time during the course of follow-up were made with the Fisher exact test.[7] In addition to the drug-placebo comparisons, the results for tolbutamide-treated patients have been compared with the results for insulin-treated patients since the primary question is which therapy should be regarded as the treatment of choice for adult-onset diabetes.

The P value obtained from the Fisher exact test represents the probability of obtaining the observed distribution or a more extreme distribution of patients with the event in each group under the null hypothesis of no difference in the occurrence of the event in the two groups being compared. However, these P values should be considered nominal P values; they provide only a measure of the relative magnitude of treatment differences, since no attempt has been made to adjust for the fact that several response variables have been considered for multiple treatment comparisons at frequent intervals during the course of the study. In our judgment, only P values less than 0.01 or perhaps 0.005 should be considered "statistically significant" in these circumstances.

In this report the results after the initiation of the assigned treatment for certain response variables are presented in terms of the per cent change from baseline for groups of patients (or cohorts) who have been followed for the same specified length of time. To be included in these analyses, a patient must have had an observation for the variable of interest at entry and at the last follow-up examination that defined the length of follow-up for the specified cohort. In addition, he/she must not have had two consecutive missing observations for the variable of interest. A missing value preceded and followed by an observed value was estimated by the mean of these two observed values.

Consideration of a specified cohort of patients has the advantage that any observed differences from one examination to another are assessed in the same group of patients. It is, of course, important that the cohorts considered should not be such a small proportion of the total number of patients in each treatment group that the results for the cohorts would not be representative of the results for the total groups.

RESULTS FROM THE HEART EXAMINATION

Specified Cardiovascular Events Identified at Follow-up Heart Examinations

A summary of the occurrence of events indicating heart disease or heart-related problems identified at follow-up heart examinations is given in part A of table 2. The percentage of patients identified during the course of follow-up with a specified nonfatal cardiac event or condition was higher in the tolbutamide-treated group than for patients in the placebo-treated group for each of the events listed in table 2 except for the occurrence of significant ECG abnormalities. The smallest P values from the Fisher exact tests of the drug-placebo comparisons were noted for the insulin standard versus placebo comparison for history of hospitalization for heart disease and the tolbutamide versus placebo comparison of use of digitalis. The tolbutamide versus insulin standard comparisons for history of hospitalization for heart disease and occurrence of angina pectoris yielded P values of 0.05. The insulin standard-treated group had fewer hospitalizations and fewer patients with angina pectoris than the placebo-treated group and the tolbutamide-treated group. More patients in the tolbutamide-treated group than in the placebo-treated group reported use of digitalis.

The percentages of patients identified with each of the specific ECG abnormalities (classified according to the Minnesota Code[8,9] by graders who had no knowledge of treatment group or clinical findings for a patient) included in the category "significant ECG abnormalities" are given in table 3. The abnormalities most frequently identified in each treatment group

TABLE 2

Events identified at follow-up heart examinations

Part A. Per cent of patients with a specified event identified at a follow-up heart examination among patients who did not have the indicated event at baseline (Number of patients with a specified event given in parentheses)

Specified event	PLBO	TOLB	ISTD	IVAR
Significant ECG abnormalities*	12.6 (24)	9.5 (18)	8.9 (17)	11.7 (22)
History of digitalis use	4.2 (8)	8.9 (16)	6.8 (13)	5.4 (10)
Hospitalization for heart disease	5.2 (10)	5.8 (11)	1.6 (3)	3.2 (6)
Angina pectoris	11.1 (21)	15.8 (29)	8.6 (16)	10.2 (19)

Part B. Denominators

Specified event	PLBO	TOLB	ISTD	IVAR
Significant ECG abnormalities*	190	190	192	188
History of digitalis use	190	180	190	184
Hospitalization for heart disease	194	190	190	187
Angina pectoris	189	183	187	187

Part C. P values from Fisher exact tests comparing the observed distribution of patients with the specified event

Specified event	TOLB versus PLBO	ISTD versus PLBO	IVAR versus PLBO	TOLB versus ISTD	TOLB versus IVAR
Significant ECG abnormalities*	0.41	0.30	0.91	0.97	0.59
History of digitalis use	0.10	0.37	0.76	0.59	0.28
Hospitalization for heart disease	0.96	0.09	0.49	0.05	0.34
Angina pectoris	0.24	0.51	0.90	0.05	0.14

*Major or moderate Q-waves (codes 1-1-1 through 1-2-7), S-T depression (code 4-1), T-wave inversion (code 5-1), complete heart block (code 6-1), left bundle branch block (code 7-1), or ventricular tachycardia (code 8-2). All ECGs analyzed by the Minnesota Code.[8,9]

were changes in Q-waves (Minnesota Codes 1-1-1 through 1-2-7, see Appendix A). Very few patients developed more than one ECG abnormality during the course of follow-up. This is evident from the fact that the sum of the percentages of the individual categories of abnormalities is only slightly greater than the percentage of patients identified with one or more of all the abnormalities listed. In the placebo-treated group, the sum of the percentages of patients with a specified abnormality was 13.1 per cent and the percentage of patients with one or more abnormalities was 12.6 per cent, indicating only one placebo-treated patient had more than one of these abnormalities. Similar results were noted for the patients in the other treatment groups. There were no patients in the tolbutamide-treated group with more than one of these abnormalities; one patient in the insulin standard treatment group and three patients in the insulin variable treatment group had more than one abnormality.

Myocardial Infarctions

The percentage of patients who reported having been hospitalized because of a myocardial infarction (MI) is given for each treatment group on the top line of part A of table 4. More patients in the placebo-treated group reported being hospitalized for a nonfatal myocardial infarction than did the patients in any of the other treatment groups.

The findings on the electrocardiograms obtained at follow-up heart examinations were compared with the ECG findings obtained at baseline for each of the patients considered in this report. Changes from a less severe finding for Q/QS and T patterns on the baseline ECG to a more severe finding for these abnormalities on a follow-up ECG were considered suggestive evi-

TABLE 3

Per cent of patients with significant ECG abnormalities identified at a follow-up heart examination for patients who did not have any of these abnormalities at baseline

(Number of patients with a specified abnormality given in parentheses)

ECG abnormality	PLBO	TOLB	ISTD	IVAR
Major or moderate Q-waves	6.8 (13)	5.3 (10)	5.2 (10)	5.9 (11)
S-T depression	4.2 (8)	2.1 (4)	2.6 (5)	5.3 (10)
T-wave inversion	0.0 (0)	0.5 (1)	0.0 (0)	1.1 (2)
Complete heart block	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)
Left bundle branch block	2.1 (4)	1.6 (3)	1.6 (3)	1.1 (2)
Ventricular tachycardia	0.0 (0)	0.0 (0)	0.0 (0)	0.0 (0)
Any of the above	12.6 (24)	9.5 (18)	8.9 (17)	11.7 (22)
Number of patients	190	190	192	188

TABLE 4

Occurrence of myocardial infarctions

Part A. Per cent of patients with a specified event indicative or suggestive of a myocardial infarction (Number of patients with a specified event given in parentheses)

Specified event	PLBO	TOLB	ISTD	IVAR
Hospitalization for nonfatal MI	3.9 (8)	2.0 (4)	1.5 (3)	0.5 (1)
Significant worsening Q/QS				
One or more examinations	5.7 (11)	4.1 (8)	5.1 (10)	6.3 (12)
Two consecutive examinations	1.1 (2)	1.7 (3)	2.7 (5)	1.1 (2)
Significant worsening Q/QS+T				
One or more examinations	3.6 (7)	4.1 (8)	2.0 (4)	2.6 (5)
Two consecutive examinations	0.5 (1)	0.6 (1)	0.5 (1)	0.6 (1)
All nonfatal MI	9.8 (20)	7.8 (16)	7.3 (15)	8.1 (16)

Part B. Denominators

Specified event	PLBO	TOLB	ISTD	IVAR
Hospitalization for nonfatal MI	204	204	206	198
Significant worsening Q/QS				
One or more examinations	194	193	197	191
Two consecutive examinations	184	181	185	180
Significant worsening Q/QS+T				
One or more examinations	194	193	202	191
Two consecutive examinations	184	181	192	179
All nonfatal MI	204	204	206	198

Part C. P values from Fisher exact tests comparing the observed distribution of patients with the specified event

Specified event	TOLB versus PLBO	ISTD versus PLBO	IVAR versus PLBO	TOLB versus ISTD	TOLB versus IVAR
Hospitalization for nonfatal MI	0.38	0.21	0.04	0.99	0.39
Significant worsening Q/QS					
One or more examinations	0.65	0.97	0.97	0.84	0.48
Two consecutive examinations	0.98	0.45	1.00	0.75	1.00
Significant worsening Q/QS+T					
One or more examinations	0.99	0.50	0.79	0.34	0.59
Two consecutive examinations	1.00	1.00	1.00	1.00	1.00
All nonfatal MI	0.60	0.46	0.67	0.98	1.00

dence that a nonfatal myocardial infarction had occurred some time after entry into the study. These changes are referred to as "significant worsening" in Q/QS or Q/QS and T patterns and are defined in Appendix A. The percentage of patients with Q/QS changes was highest in the insulin variable treatment group (6.3) and lowest in the tolbutamide treatment group (4.1). The percentages of patients with significant worsening of Q/QS and T patterns were almost the same in all four treatment groups, ranging from 2.0 for the insulin standard group to 4.1 for the tolbutamide group (table 4).

The persistence of these changes was determined by considering the number of patients who had these ECG changes noted on two consecutive ECGs obtained during follow-up. Forty-one patients were identified with Q/QS changes, and 12 of these had a second ECG with these same Q/QS changes. Among the 24 patients with Q/QS-plus-T-pattern changes at one examination, four had significant worsening observed on the ECG obtained at the next annual heart examination (table 4).

Reports of hospitalization for myocardial infarctions and significant worsening in the findings of the follow-up electrocardiograms were considered indicators that a patient had sustained a nonfatal myocardial infarction. The percentage of patients with one or more of these findings was slightly higher in the placebo-treated group than in the other three treatment groups.

The drug-placebo comparisons as well as the TOLB-ISTD and TOLB-IVAR comparisons for each of the events listed in table 4 were made with the Fisher exact test. The P values for these tests are given in part C of table 4. The smallest P value (0.04) was observed for the IVAR-PLBO comparison of hospitalizations for nonfatal MI.

The number of patients who had a nonfatal myocardial infarction (hospitalized for myocardial infarction or significant worsening of Q/QS or Q/QS and T patterns) during the course of follow-up is given for each treatment group on the second line of table 5. Fifty-one of these 67 patients were alive as of October 31, 1969. Five of these patients were reported to have

died of myocardial infarction or sudden death as of October 31, 1969, and the remainder (11 patients) had died of other causes. A "sudden death" was defined as death occurring within three hours of the onset of symptoms in an otherwise clinically stable patient and in a manner consistent with a cardiovascular catastrophe. A death was classified as a fatal myocardial infarction if this diagnosis was made from ECG changes and changes in serum enzymes observed during the terminal course of illness or if the events leading to death were clinically compatible with this diagnosis and autopsy findings provided evidence that a myocardial infarction was the principal cause of death. The number of patients alive and the number dead due to myocardial infarction or sudden death among the patients identified with a nonfatal myocardial infarction are given for each treatment group on the third and fourth lines of table 5.

Patients who were hospitalized for a myocardial infarction and died or who died of a sudden death but who had not had a nonfatal myocardial infarction (as defined above) prior to the fatal event were classified in the category "fatal MI only." The number of these patients is given for each treatment group on the fifth line of table 5. The results for fatal and nonfatal myocardial infarctions have been pooled since the event is the same; only the outcome is different. The total number of patients who had either a fatal myocardial infarction, nonfatal myocardial infarction, or both is given on the sixth line of table 5. The percentage of patients identified with fatal or nonfatal myocardial infarction was approximately the same in all treatment groups (10.7 for PLBO, 13.7 for TOLB, 11.0 for ISTD, and 12.3 for IVAR).

The per cent mortality or case fatality for all patients who sustained a myocardial infarction may be determined on the basis of the information presented in table 5. For example, the mortality in the placebo treatment group was 18.2 per cent $(2+2/22)\times 100$. The mortality in the other groups was 50.0 per cent for TOLB, 34.8 per cent for ISTD, and 40.0 per cent for IVAR. Although the occurrence of fatal and nonfatal myocardial infarctions was not different among the treatment groups, these results suggest that myocardial infarctions were more likely to be fatal for patients in the tolbutamide-treated group.

Use of Medications for Heart Disease or Heart-related Problems

The percentages of patients who reported taking certain medications for heart disease at one or more follow-up examinations during the course of the study are given for each treatment group in part A of table 6. Information on use of digitalis and history of angina pectoris was obtained for each patient during the baseline heart examination, and patients identified with a history of digitalis use or a history of angina pectoris were not considered at risk of use of digitalis or of antianginal drugs, respectively, in the analyses presented here. No information on use of other drugs was collected during the baseline examination; therefore, all patients who completed at least one quarterly follow-up examination were considered at risk of using one of the drugs considered in the analyses of other drugs for heart disease or heart-related problems.

The largest differences among the treatment groups were observed for the use of digitalis or drugs for congestive heart failure; fewer patients in the

TABLE 5

Per cent of patients with nonfatal and fatal myocardial infarctions

Part A. Per cent of patients with fatal and/or nonfatal myocardial infarctions (Number of patients with a specified event given in parentheses)

	PLBO	TOLB	ISTD	IVAR
Number of patients at risk	205	204	210	204
Specified event				
Nonfatal MI	9.8 (20)	7.8 (16)	7.1 (15)	7.8 (16)
Status 10/31/69				
Alive	8.3 (17)	5.4 (11)	4.8 (10)	6.4 (13)
Dead because of MI or sudden death	1.0 (2)	1.0 (2)	0.0 (0)	0.5 (1)
Fatal MI only	1.0 (2)	5.9 (12)	3.8 (8)	4.4 (9)
Nonfatal or fatal MI	10.7 (22)	13.7 (28)	11.0 (23)	12.3 (25)
Case fatality in per cent	18.2	50.0	34.8	40.0

Part B. P values from Fisher exact tests comparing the observed distribution of patients with the specified event

Specified event	TOLB versus PLBO	ISTD versus PLBO	IVAR versus PLBO	TOLB versus ISTD	TOLB versus IVAR
Nonfatal MI	0.61	0.44	0.61	0.99	0.99
Fatal MI only	0.01	0.11	0.06	0.43	0.12
Nonfatal or fatal MI	0.44	1.00	0.74	0.53	0.42

TABLE 6

Medication for heart disease or heart-related problems

Part A. Per cent of patients who completed at least one quarterly follow-up examination and who reported taking the indicated medication or taking a medication for the indicated reason at one or more of these examinations (Number of patients taking a specified medication given in parentheses)

Drug or reason	PLBO	TOLB	ISTD	IVAR
Digitalis glycosides or congestive heart failure	6.7 (13)	11.6 (22)	12.4 (24)	9.6 (18)
Number of patients*	195	189	194	188
Antianginal agents or angina pectoris	6.2 (12)	6.3 (12)	6.8 (13)	5.2 (10)
Number of patients†	194	190	190	191
Antihypertensive agents or hypertension	23.5 (48)	27.0 (55)	30.1 (62)	24.7 (49)
Antihypertensive-diuretics‡	15.2 (31)	16.2 (33)	17.5 (36)	15.2 (30)
Either of above	23.5 (48)	27.0 (55)	30.6 (63)	25.8 (51)
Antiarrhythmic agents or arrhythmias	3.9 (8)	4.9 (10)	3.9 (8)	4.0 (8)
Diuretics	28.4 (58)	31.4 (64)	31.1 (64)	24.2 (48)
Anticoagulants	3.4 (7)	2.0 (4)	3.4 (7)	1.0 (2)
ASHD or coronary insufficiency	6.4 (13)	8.3 (17)	8.3 (17)	6.6 (13)
Other heart disease	0.5 (1)	1.5 (3)	1.9 (4)	0.0 (0)
Nonspecific problems	6.9 (14)	5.9 (12)	7.8 (16)	5.1 (10)
Number of patients	204	204	206	198

Part B. P values from Fisher exact tests comparing the observed distribution of patients who completed at least one quarterly follow-up examination and who reported taking the indicated medication or taking a medication for the indicated reason at one or more of these examinations

Drug or reason	TOLB versus PLBO	ISTD versus PLBO	IVAR versus PLBO	TOLB versus ISTD	TOLB versus IVAR
Digitalis glycosides or congestive heart failure	0.13	0.08	0.39	0.95	0.63
Antianginal agents or angina pectoris	1.00	0.96	0.86	1.00	0.82
Antihypertensive agents or hypertension	0.49	0.16	0.87	0.55	0.70
Antihypertensive-diuretics‡	0.89	0.62	1.00	0.83	0.88
Either of above	0.49	0.13	0.69	0.48	0.87
Antiarrhythmic agents or arrhythmias	0.81	1.00	1.00	0.79	0.86
Diuretics	0.59	0.63	0.40	1.00	0.14
Anticoagulants	0.54	1.00	0.19	0.55	0.71
ASHD or coronary insufficiency	0.57	0.59	1.00	1.00	0.63
Other heart disease	0.62	0.38	1.00	1.00	0.26
Nonspecific heart problems	0.84	0.87	0.58	0.58	0.88

*Excludes patients who reported use of digitalis at the baseline heart examination.
†Excludes patients who reported a history of angina pectoris at the baseline heart examination.
‡Combination antihypertensive-diuretic agents.[5]

placebo-treated group than in any of the drug-treated groups reported taking digitalis or drugs for congestive heart failure. The Fisher exact test of the number of patients who reported taking digitalis and/or a drug for congestive heart failure at one or more follow-up examinations for the tolbutamide-placebo comparison and the insulin standard-placebo comparison yielded P values of 0.13 and 0.08, respectively.

Blood Pressure Levels During the Course of Follow-up

The per cent change from baseline in mean systolic blood pressure levels at each follow-up heart examination is plotted in figure 1 for the cohort of patients who were followed through the 18th follow-up examination. The mean systolic blood pressure level at entry in the cohorts of patients studied was 145 mm. Hg for PLBO, 143 mm. Hg for TOLB, 145 mm. Hg for ISTD, and 141 mm. Hg for IVAR. A slight decrease in systolic blood pressure was noted during the early period of follow-up. This initial decrease was followed by a rather constant but insignificant increase in all treatment groups.

The per cent change in mean diastolic blood pres-

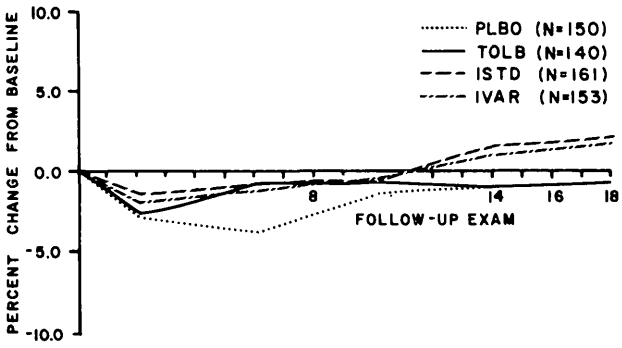


FIG. 1. Per cent change in systolic blood pressure from baseline to each follow-up examination for the cohort of patients followed through the 18th follow-up examination.

sure levels for the cohort of patients followed through the 18th follow-up examination is plotted for each treatment group in figure 2. The mean diastolic blood pressure level at baseline for each of these cohorts was 86 mm. Hg for PLBO, 84 mm. Hg for TOLB, 86 mm. Hg for ISTD, and 83 mm. Hg for IVAR. Although there were small increases in diastolic blood pressure levels after the initial decrease in all treatment groups, the mean levels remained below the baseline levels throughout the period of observation.

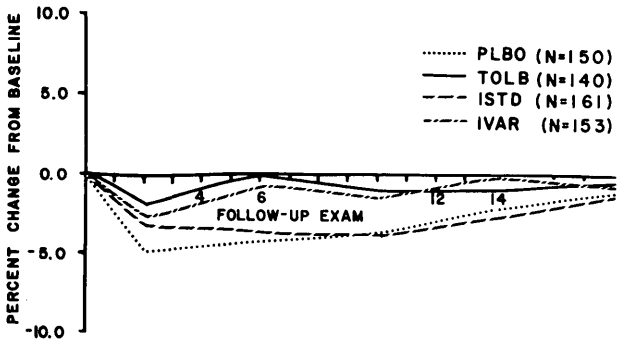


FIG. 2. Per cent change in diastolic blood pressure from baseline to each follow-up examination for the cohort of patients followed through the 18th follow-up examination.

Occurrence of Hypertension and Treatment for It

The assessment of the development of hypertension during the course of the study was based on the occurrence of elevated blood pressure levels and the reporting of treatment for hypertension. These results are given in table 7. For this analysis, hypertension was defined according to the World Health Organization criteria:[10] systolic blood pressure ≥ 160 mm. Hg and/or diastolic blood pressure ≥ 95 mm. Hg. The percentage of patients who had elevated blood pressure levels indicative of hypertension at one or more follow-up heart examinations and at two or more follow-up heart examinations is given for each treatment group in the first lines of parts A and B, respectively, of table 7.

The percentages of patients who were treated for hypertension at least once or at least twice during the course of follow-up among those patients who were not classified as hypertensive at baseline are given for each treatment group in the second lines of parts A

TABLE 7

Per cent of patients classified as hypertensive* and/or treated for hypertension during the course of follow-up among patients who were not classified as hypertensive* at baseline

(Number of patients with a specified event given in parentheses)

	PLBO	TOLB	ISTD	IVAR
Part A. Patients who completed at least one follow-up heart examination				
Identified at one or more follow-up examinations:				
Elevated blood pressure levels indicative of hypertension	39.8 (51)	44.5 (61)	41.7 (58)	45.1 (64)
Treated for hypertension	9.4 (12)	16.8 (23)	18.7 (26)	17.6 (25)
Both of the above	7.8 (10)	11.7 (16)	12.9 (18)	14.1 (20)
Number of patients	128	137	139	142
Part B. Patients who completed at least two follow-up examinations				
Identified at two or more follow-up examinations:				
Elevated blood pressure levels indicative of hypertension	24.2 (30)	21.9 (28)	29.8 (39)	28.5 (39)
Treated for hypertension	6.5 (8)	12.5 (16)	11.5 (15)	13.1 (18)
Both of the above	5.6 (7)	7.8 (10)	9.2 (12)	9.5 (13)
Number of patients	124	128	131	137

Part C. P values from Fisher exact tests comparing the observed distribution of patients with the specified event

	TOLB versus PLBO	ISTD versus PLBO	IVAR versus PLBO	TOLB versus ISTD	TOLB versus IVAR
Identified at one or more follow-up examinations:					
Elevated blood pressure levels indicative of hypertension	0.52	0.85	0.46	0.73	1.00
Treated for hypertension	0.11	0.04	0.07	0.80	0.98
Both of the above	0.40	0.24	0.15	0.89	0.68
Identified at two or more follow-up examinations:					
Elevated blood pressure levels indicative of hypertension	0.77	0.39	0.52	0.19	0.28
Treated for hypertension	0.15	0.24	0.11	0.94	1.00
Both of the above	0.67	0.41	0.35	0.87	0.79

*Systolic blood pressure ≥ 160 mm. Hg and/or diastolic blood pressure ≥ 95 mm. Hg.

and B of table 7. A patient was identified as receiving treatment for hypertension if at one or more quarterly follow-up examinations the patient reported he/she had been given a prescription for a specific antihypertensive drug or drugs‡ or if the patient reported taking a medication for the treatment of hypertension even if the drug was not identified.

A smaller percentage of patients in the placebo-treated group than in the insulin-treated groups had elevated blood pressure levels or reported receiving treatment for hypertension whether the analysis was based on the occurrence at one or more examinations or at two or more examinations. The P values from the Fisher exact tests are given in part C of table 7. P values < 0.10 were obtained for the PLBO-ISTD and PLBO-IVAR comparisons of treatment for hypertension at one or more examinations.

Review of the blood pressure readings for patients in the placebo and tolbutamide group who did not have elevated blood pressure levels at baseline or other indications of cardiovascular disease and who had at least three follow-up heart examinations also indicated no tolbutamide-placebo differences. Elevation of both systolic and diastolic blood pressure was observed in 12 patients in the placebo-treated group and 19 patients in the tolbutamide-treated group. Only 13 patients in each of these two groups ever had a systolic blood pressure reading greater than 180 mm. Hg; 14 patients in the placebo group and 20 in the tolbutamide group had one or more diastolic blood pressure readings greater than 100 mm. Hg.

Heart Rate

The mean heart rate (read from the ECG) at baseline for the cohort of patients completing the 18th follow-up examination was 70 beats per minute for PLBO, 70 beats per minute for TOLB, 71 beats per minute for ISTD, and 72 beats per minute for IVAR. All changes in heart rate in this cohort during the course of follow-up were within 2 per cent of the baseline levels (figure 3).

Cardiothoracic Ratio

The mean cardiothoracic ratio at baseline for patients in the cohort followed through the 20th follow-up examination was approximately equal in all treatment groups (0.48 for PLBO, 0.47 for TOLB, 0.48 for ISTD, and 0.48 for IVAR). The changes from baseline in all treatment groups were 2 per cent to 3 per cent above baseline levels during the final periods of observation with the exception of patients in the placebo-treated group who showed even less change from baseline (figure 4).

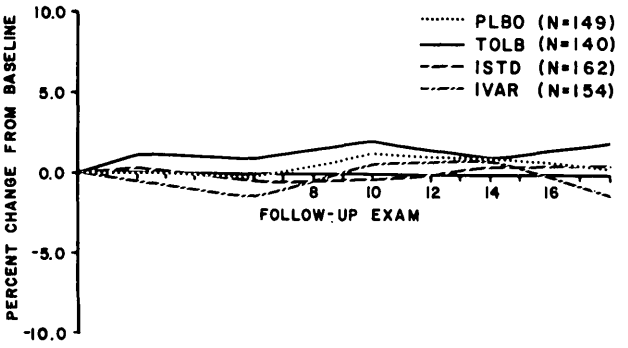


FIG. 3. Per cent change in heart rate from baseline to each follow-up examination for the cohort of patients followed through the 18th follow-up examination.

RESULTS FROM OTHER EXAMINATIONS

Findings Obtained During Follow-up Eye Examinations

The percentage of patients identified with an eye condition during the course of follow-up is given for each treatment group in part A of table 8. Each of the drug-placebo comparisons for the conditions listed in table 8 was obtained with the Fisher exact test. The smallest P value (0.13) was obtained for the insulin standard versus placebo comparison of the number of patients identified with exudates in the fundus photographs; more patients in the insulin standard-treated group had evidence of exudates than did patients in the placebo-treated group. Exudates were considered

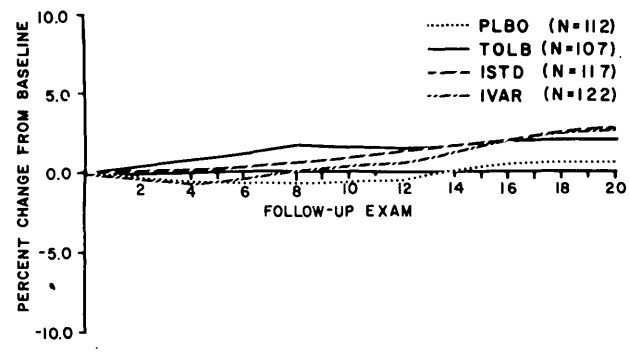


FIG. 4. Per cent change in cardiothoracic ratio from baseline to each follow-up examination for the cohort of patients followed through the 20th follow-up examination.

‡Combination antihypertensive and diuretic drugs such as Aldoril, Hydromox, or Serpasil were regarded as specific antihypertensive drugs. Diuretic drugs such as chlorothiazide and hydrochlorothiazide were not considered antihypertensive agents in these tabulations unless the reported reason the drug was prescribed was treatment of hypertension.

TABLE 8

Events identified at follow-up eye examinations

Part A. Per cent of patients with a specified event identified at a follow-up eye examination among patients who did not have the indicated event at baseline (Number of patients with a specified event given in parentheses)

Specified event	PLBO	TOLB	ISTD	IVAR
Visual acuity ≤ 20/200 (either eye)	6.7 (12)	9.4 (17)	5.6 (10)	5.7 (10)
Visual acuity ≤ 20/200 (both eyes)	2.7 (5)	3.7 (7)	1.6 (3)	3.8 (7)
Hospitalization for eye disease	3.9 (7)	1.6 (3)	3.3 (6)	3.4 (6)
Opacity (vitreous, lenticular, or corneal—either eye)	7.5 (13)	10.5 (19)	7.3 (13)	8.7 (15)
Fundus abnormalities excluding exudates*	42.9 (54)	41.7 (50)	44.4 (52)	39.8 (47)
Exudates	51.2 (43)	53.5 (53)	64.7 (44)	48.6 (36)

Part B. Denominators

Specified event	PLBO	TOLB	ISTD	IVAR
Visual acuity ≤ 20/200 (either eye)	179	181	179	175
Visual acuity ≤ 20/200 (both eyes)	186	189	188	184
Hospitalization for eye disease	178	185	182	179
Opacity (vitreous, lenticular, or corneal—either eye)	173	181	179	173
Fundus abnormalities excluding exudates*	126	120	117	118
Exudates†	84	99	68	74

Part C. P values from Fisher exact tests comparing the observed distribution of patients with the specified event

Specified event	TOLB versus PLBO	ISTD versus PLBO	IVAR versus PLBO	TOLB versus ISTD	TOLB versus IVAR
Visual acuity ≤ 20/200 (either eye)	0.46	0.83	0.87	0.24	0.27
Visual acuity ≤ 20/200 (both eyes)	0.79	0.71	0.76	0.34	1.00
Hospitalization for eye disease	0.31	0.97	0.99	0.49	0.47
Opacity (vitreous, lenticular, or corneal—either eye)	0.43	1.00	0.84	0.37	0.69
Fundus abnormalities excluding exudates*	0.95	0.90	0.73	0.76	0.88
Exudates†	0.87	0.13	0.87	0.20	0.65

*Readings of right central fundus photographs for one or more of the following abnormalities: retinal hemorrhages and microaneurysms, preretinal and vitreous hemorrhages, venous pathology, arterial pathology, or proliferative changes and neovascularization.

†Based on readings of right central fundus photographs.

separately since it was not always possible for the reader to differentiate true exudates from artifacts appearing on the film or from drusen bodies found in normal retinae. None of the other comparisons yielded P values less than 0.20 (part C of table 8).

The specific abnormalities, excluding exudates, that were identified in the right central fundus photographs are given in part A of table 9. Other than exudates, the findings that were observed most frequently in the fundus photographs were retinal hemorrhages and microaneurysms. Proliferative changes and neovascularization based on the grading of the central right photograph were identified in only two patients, one patient in the placebo treatment group and one in the insulin standard treatment group.

Fundus photographs were not obtained for all patients at the baseline examination because of technical problems with the Zeiss camera. Therefore, an analysis was also performed in which either the baseline photographs or those taken at quarter 1 (three months after entry) were used to define the baseline status for each patient. In this analysis the identification of the new findings during the course of follow-up was based on all follow-up eye examinations excluding the results from the eye examination completed at quarter 1. The results of this analysis are summarized in part B of table 9, and they are similar to the results obtained when only baseline findings were used to classify the patient's status at baseline.

Nonfatal Events Identified at Follow-up Kidney Examinations

The mean serum creatinine level at baseline was 0.91 mg. per 100 ml. for PLBO, 0.90 mg. per 100 ml. for TOLB, 0.87 mg. per 100 ml. for ISTD, and 0.91 mg. per 100 ml. for IVAR for patients followed through the 19th follow-up examination. Only small changes from the baseline levels were noted in all treatment groups during this period of observation (figure 5).

The percentage of patients classified with a specified condition defined on the basis of the various kidney function tests is given for each of the treatment groups in part A of table 10. The smallest P value

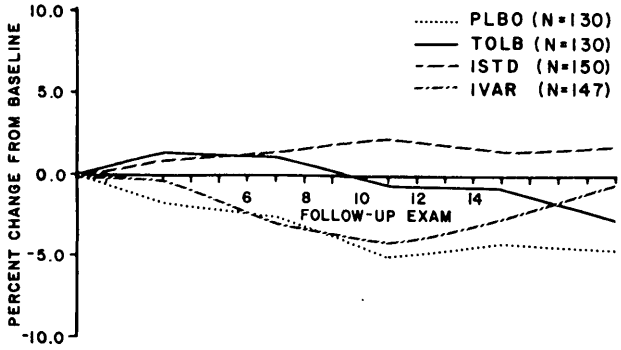


FIG. 5. Per cent change in serum creatinine levels from baseline to each follow-up examination for the cohort of patients followed through the 19th follow-up examination.

(0.06) was obtained for the TOLB versus ISTD comparison for urine protein levels ≥ 1 gm. per L. None of the patients in the insulin standard group had urine protein levels ≥ 1 gm. per L. in the time period covered by this analysis.

Findings Obtained During Follow-up

Peripheral Vascular Examinations

A summary of conditions identified at follow-up peripheral vascular examinations is given in part A of table 11. Two patients in the placebo-treated group and one patient in the insulin variable-treated group had an amputation of a lower limb during the course of follow-up. The reported reason for the amputation was gangrene in every instance. The percentage of patients identified with arterial calcification was highest in the insulin variable-treated group and lowest in the placebo-treated group. More patients were identified with intermittent claudication in the tolbutamide-treated group than in any of the other treatment groups. The TOLB-PLBO comparison for this event yielded the smallest P value (0.08).

Body Weight

The changes in body weight for patients in each of the treatment groups considered in this report are presented in figure 6. For this analysis, changes in body weight have been evaluated in terms of relative body weight, which was defined as the ratio of a patient's observed body weight to his/her desirable body weight. Desirable body weight was derived from tables published by the Metropolitan Life Insurance Company.[11]

Patients in all treatment groups at the time of entry into the study were on the average 34 per cent overweight. After the initiation of treatment, which included a prescription for diet with caloric content designed to achieve or to maintain the patient's body weight within 15 per cent of the desirable amount, there was a small initial loss of weight for patients in all treatment groups. The initial weight loss was maintained only for patients in the placebo-treated group.

Serum Cholesterol Levels

The per cent change in serum cholesterol levels from baseline to each follow-up examination is presented in figure 7 for the cohort of patients in each treatment group followed through the 20th follow-up

TABLE 9

Per cent of patients with a specified fundus abnormality identified at a follow-up eye examination for patients who did not have any of these abnormalities at baseline

Fundus abnormality	(Number of patients with a specified abnormality given in parentheses) PLBO	TOLB	ISTD	IVAR
Part A. Patients with photographs taken at baseline				
Retinal hemorrhages and microaneurysms	39.7 (50)	40.8 (49)	42.7 (50)	36.4 (43)
Preretinal and vitreous hemorrhages	2.4 (3)	1.7 (2)	1.7 (2)	2.5 (3)
Venous pathology	3.2 (4)	1.7 (2)	2.6 (3)	6.8 (8)
Arterial pathology	0.0 (0)	0.8 (1)	0.9 (1)	0.0 (0)
Proliferative changes and neovascularization	0.8 (1)	0.0 (0)	0.9 (1)	0.0 (0)
Any of the above	42.9 (54)	41.7 (50)	44.4 (52)	39.8 (47)
Number of patients	126	120	117	118
Part B. Patients with photographs taken at baseline or at first follow-up eye examination*				
Retinal hemorrhages and microaneurysms	39.7 (62)	34.9 (52)	35.1 (54)	29.4 (42)
Preretinal and vitreous hemorrhages	1.9 (3)	2.0 (3)	1.9 (3)	3.5 (5)
Venous pathology	2.6 (4)	2.7 (4)	4.5 (7)	4.9 (7)
Arterial pathology	0.0 (0)	0.7 (1)	1.3 (2)	0.0 (0)
Proliferative changes and neovascularization	0.6 (1)	0.7 (1)	0.6 (1)	0.7 (1)
Any of the above	42.9 (67)	36.9 (55)	38.3 (59)	32.9 (47)
Number of patients	156	149	154	143

*Photographs from first follow-up eye examination used to define baseline status for the specified event if photographs were not taken at baseline.

TABLE 10

Events identified at follow-up kidney examinations

Part A. Per cent of patients with a specified event identified at a follow-up kidney examination among patients who did not have the indicated event at baseline (Number of patients with a specified event given in parentheses)

Specified event	PLBO	TOLB	ISTD	IVAR
Serum creatinine ≥ 1.5 mg./100 ml.	5.4 (10)	4.8 (9)	3.6 (7)	4.8 (9)
Corrected creatinine clearance ≤ 50 ml./min.	13.8 (25)	11.5 (21)	14.7 (28)	8.8 (16)
Hospitalization for kidney disease	0.6 (1)	1.1 (2)	2.1 (4)	2.7 (5)
Urine protein ≥ 1 gm./L.	2.1 (4)	2.6 (5)	0.0 (0)	3.7 (7)
Urine protein 2+ or greater at two consecutive quarterly follow-up examinations	4.8 (9)	5.3 (10)	6.3 (12)	6.4 (12)

Part B. Denominators

Specified event	PLBO	TOLB	ISTD	IVAR
Serum creatinine ≥ 1.5 mg./100 ml.	184	189	195	187
Corrected creatinine clearance ≤ 50 ml./min.	181	182	191	182
Hospitalization for kidney disease	180	180	188	183
Urine protein ≥ 1 gm./L.	189	191	194	190
Urine protein 2+ or greater at two consecutive quarterly follow-up examinations	186	189	190	187

Part C. P values from Fisher exact tests comparing the observed distribution of patients with the specified event

Specified event	TOLB versus PLBO	ISTD versus PLBO	IVAR versus PLBO	TOLB versus ISTD	TOLB versus IVAR
Serum creatinine ≥ 1.5 mg./100 ml.	0.95	0.54	0.97	0.75	1.00
Corrected creatinine clearance ≤ 50 ml./min.	0.62	0.93	0.18	0.46	0.49
Hospitalization for kidney disease	1.00	0.40	0.22	0.73	0.46
Urine protein ≥ 1 gm./L.	1.00	0.12	0.55	0.06	0.76
Urine protein 2+ or greater at two consecutive quarterly follow-up examinations	1.00	0.69	0.66	0.84	0.81

examination. The mean serum cholesterol levels at baseline for patients followed through the 20th follow-up examination was 229 mg. per 100 ml. for PLBO, 236 mg. per 100 ml. for TOLB, 240 mg. per 100 ml. for ISTD, and 241 mg. per 100 ml. for IVAR. A decrease of 3.3 per cent during the first year of follow-up was noted for patients in the insulin variable-treated group whereas there was an increase of 3.0 per cent for patients in the placebo-treated group. After five years of follow-up (i.e., at the 20th follow-up examination) there was an increase of 7.7 per cent in the placebo-treated group; smaller increases were noted for patients in the tolbutamide- and insulin standard-treated groups, and the mean value for the insulin variable-treated patients had just returned to the baseline level.

Use of Other Medications

The percentages of patients who reported taking specific types of drugs during the course of follow-up are given in table 12. The findings for the use of drugs for heart disease or heart-related problems were presented in table 6.

More patients in the placebo-treated group than in the insulin standard-treated group reported taking erythromycin or derivatives and aspirin-containing analgesics (P values for these comparisons were

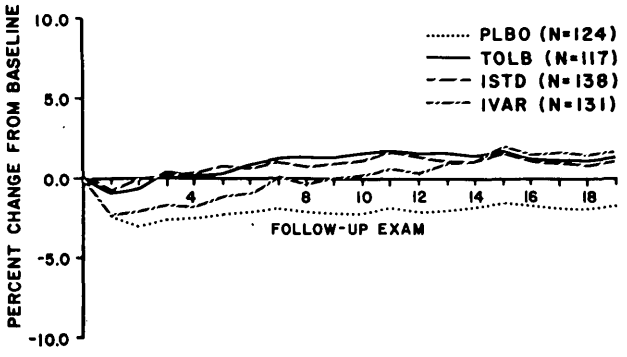


FIG. 6. Per cent change in relative body weight from baseline to each follow-up examination for the cohort of patients followed through the 19th follow-up examination.

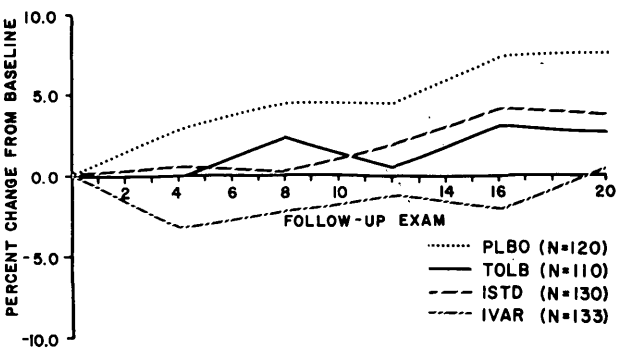


FIG. 7. Per cent change in serum cholesterol levels from baseline to each follow-up examination for the cohort of patients followed through the 20th follow-up examination.

< 0.05, indicated by daggers in table 12). More patients in the insulin standard-treated group than in the placebo-treated group reported taking adrenal corticosteroids and corticotropin (P value < 0.05). Four of the tests of TOLB versus PLBO comparisons yielded P values < 0.10 but ≥ 0.05 (indicated by an asterisk in table 12). These P values were obtained from the comparisons of agents used in gout, antihistamines, cold remedies, and peripheral vasodilators, and, except for the use of antihistamines, more patients in the tolbutamide treatment group than in the placebo treatment group reported taking the indicated medication.

An analysis of the reported reasons for taking specified drugs is given in table 13. The largest differences noted for the TOLB-versus-PLBO comparisons were in the reported use of drugs taken for gastrointestinal infections and of those taken for unspecified thyroid maladies. A larger percentage of patients in the tolbutamide group reported taking medication for unspecified thyroid maladies, but a smaller percentage in this group reported taking medication for gastrointestinal infections. The Fisher exact tests of these comparisons yielded P values < 0.10 but ≥ 0.05. The Fisher exact test for the ISTD versus PLBO comparison of the reported use of drugs for trauma and environmental exposure (7.4 per cent in PLBO and 2.9 per cent in ISTD) yielded a P value < 0.10 but ≥ 0.05. Patients in the insulin variable-treated group reported taking fewer medications for abscesses and inflammations and for female genital infections than patients in the placebo-treated group (P value < 0.10 but ≥ 0.05 from the Fisher exact tests of both of these comparisons).

Cancer

Information on the occurrence of cancer was obtained by reviewing the information on hospitalizations recorded on the quarterly follow-up examination

TABLE 11

Events identified at follow-up peripheral vascular examinations

Part A. Per cent of patients with a specified event identified at a follow-up peripheral vascular examination among patients who did not have the indicated event at baseline (Number of patients with a specified event given in parentheses)

Specified event	PLBO	TOLB	ISTD	IVAR
Amputation of all or part of lower limb				
Right side	0.5 (1)	0.0 (0)	0.0 (0)	0.5 (1)
Left side	0.5 (1)	0.0 (0)	0.0 (0)	0.0 (0)
Either side	1.0 (2)	0.0 (0)	0.0 (0)	0.5 (1)
Arterial calcification*	13.1 (22)	15.6 (24)	17.9 (29)	18.8 (29)
Intermittent claudication	8.8 (16)	15.4 (28)	11.1 (21)	10.0 (18)
Absence of arterial pulses				
One or more pulses	27.3 (48)	21.7 (38)	23.0 (43)	22.4 (38)
Two or more pulses	20.5 (36)	15.4 (27)	19.3 (36)	14.7 (25)

Part B. Denominators

Specified event	PLBO	TOLB	ISTD	IVAR
Amputation of all or part of lower limb				
Right side	194	194	197	190
Left side	194	194	197	189
Either side	194	194	197	189
Arterial calcification*	168	154	162	154
Intermittent claudication	182	182	190	180
Absence of arterial pulses				
One or more pulses	176	175	187	170
Two or more pulses	176	175	187	170

Part C. P values from Fisher exact tests comparing the observed distribution of patients with the specified event

Specified event	TOLB versus PLBO	ISTD versus PLBO	IVAR versus PLBO	TOLB versus ISTD	TOLB versus IVAR
Amputation of all or part of lower limb					
Right side	1.00	0.99	1.00	1.00	0.99
Left side	1.00	0.99	1.00	1.00	1.00
Either side	0.50	0.49	1.00	1.00	0.99
Arterial calcification*	0.63	0.29	0.21	0.69	0.55
Intermittent claudication	0.08	0.58	0.83	0.28	0.17
Absence of arterial pulses					
One or more pulses	0.28	0.41	0.35	0.87	0.99
Two or more pulses	0.28	0.88	0.21	0.41	0.97

*Evidence of arterial calcification noted in both of two independent readings of soft-tissue x-ray films of the right lower limb.

TABLE 12

Per cent of patients for whom the indicated medication was prescribed during one or more follow-up periods

Drug	PLBO (Numerators given in parentheses)	TOLB	ISTD	IVAR
Part A. Antibiotics and treatment for infections				
Cephalosporins	0.5 (1)	0.5 (1)	0.5 (1)	0.0 (0)
Chloramphenicol and derivatives	2.0 (4)	2.9 (6)	3.9 (8)	3.0 (6)
Colistins and polymyxins	1.0 (2)	0.0 (0)	0.0 (0)	0.0 (0)
Erythromycin and derivatives	8.8 (18)	4.4 (9)	2.4 (5)†	4.5 (9)
Penicillins	28.9 (59)	27.5 (56)	30.6 (63)	24.7 (49)
Sulfonamides	17.2 (35)	15.7 (32)	14.1 (29)	16.7 (33)
Tetracyclines	19.6 (40)	21.6 (44)	17.0 (35)	15.2 (30)
Any of the above	46.6 (95)	48.5 (99)	44.7 (92)	44.4 (88)
Miscellaneous antiinfectious agents				
Antifungal agents	3.9 (8)	3.4 (7)	1.9 (4)	1.0 (2)
Antimalarial agents	1.5 (3)	0.5 (1)	0.5 (1)	1.0 (2)
Antitrichomonals	0.5 (1)	2.9 (6)	0.5 (1)	0.0 (0)
Nitrofurans	2.5 (5)	2.9 (6)	4.4 (9)	4.5 (9)
Miscellaneous antibacterial agents	6.4 (13)	5.9 (12)	4.4 (9)	5.1 (10)
Part B. Analgesics and antirheumatic drugs				
Agents used in gout	5.9 (12)	11.8 (24)*	7.3 (15)	9.6 (19)
Antirheumatic agents	2.9 (6)	1.0 (2)	1.9 (4)	3.5 (7)
Local anesthetics	1.5 (3)	1.5 (3)	0.5 (1)	1.0 (2)
Mild analgesics				
Aspirin-containing	49.0 (100)	50.0 (102)	38.3 (79)†	48.0 (95)
Pure aspirin	77.5 (158)	72.1 (147)	73.3 (151)	75.3 (149)
Strong analgesics	7.4 (15)	6.9 (14)	7.8 (16)	6.1 (12)
Part C. Cold remedies				
Antihistamines	21.1 (43)	14.2 (29)*	19.4 (40)	18.2 (36)
Antitussive agents	24.5 (50)	23.5 (48)	28.2 (58)	27.8 (55)
Bronchodilators	7.8 (16)	5.4 (11)	5.8 (12)	4.5 (9)
Cold remedies	13.2 (27)	20.1 (41)*	14.6 (30)	12.1 (24)
Expectorants and inhalants	12.7 (26)	10.3 (21)	10.7 (22)	9.6 (19)
Nasal decongestants	3.4 (7)	2.9 (6)	4.9 (10)	2.5 (5)
Part D. Gastrointestinal agents				
Antacids	21.1 (43)	18.1 (37)	17.5 (36)	17.2 (34)
Antidiarrheals	3.4 (7)	4.9 (10)	3.9 (8)	5.1 (10)
Antiemetics	8.3 (17)	6.4 (13)	8.3 (17)	9.1 (18)
Antispasmodics	18.6 (38)	18.1 (37)	17.0 (35)	17.2 (34)
Laxatives and agents affecting fecal consistency	20.6 (42)	23.0 (47)	23.8 (49)	21.7 (43)
Miscellaneous gastrointestinal agents	1.0 (2)	1.0 (2)	0.0 (0)	0.5 (1)
Part E. Hormones				
Adrenal corticosteroids and corticotropin	3.9 (8)	4.9 (10)	9.2 (19)†	7.6 (15)
Estrogenes, progestogens, oral contraceptives, and ovulatory agents	8.8 (18)	10.3 (21)	8.3 (17)	9.1 (18)
Thyroid hormones and antithyroid agents	2.9 (6)	2.5 (5)	3.9 (8)	2.5 (5)
Part F. Psychoactive Drugs				
Anorexiants	7.8 (16)	10.3 (21)	11.7 (24)	12.6 (25)
Antianxiety agents	20.1 (41)	20.6 (42)	17.0 (35)	21.7 (43)
Antidepressants	3.4 (7)	2.5 (5)	4.4 (9)	3.5 (7)
Antipsychotic agents	3.9 (8)	3.9 (8)	4.9 (10)	5.6 (11)
Centrally acting skeletal muscle relaxants	2.5 (5)	4.9 (10)	3.9 (8)	2.5 (5)
Sedatives and hypnotics	30.9 (63)	29.9 (61)	35.0 (72)	33.3 (66)
Part G. Other Medications				
Agents used to treat anemia	8.8 (18)	7.8 (16)	7.3 (15)	7.6 (15)
Agents used to treat glaucoma	2.9 (6)	3.4 (7)	1.9 (4)	3.0 (6)
Antiseptics and disinfectants	0.0 (0)	1.0 (2)	1.0 (2)	1.0 (2)
Dermatologic agents	2.9 (6)	1.0 (2)	2.9 (6)	4.0 (8)
Enzymes used as drugs	1.5 (3)	0.0 (0)	0.5 (1)	1.0 (2)
Immunologic agents	4.4 (9)	2.5 (5)	3.9 (8)	3.0 (6)
Ophthalmic preparations	2.5 (5)	2.0 (4)	1.9 (4)	2.0 (4)
Peripheral vasodilators	0.5 (1)	3.4 (7)*	1.5 (3)	1.5 (3)
Replenishers and regulators of water, electrolytes, and nutrients	5.9 (12)	6.4 (13)	7.8 (16)	6.6 (13)
Topical otic preparations	1.0 (2)	0.5 (1)	1.0 (2)	0.5 (1)
Vitamins and source of vitamins	5.9 (12)	5.9 (12)	6.3 (13)	5.6 (11)
Miscellaneous drugs	2.0 (4)	2.5 (5)	2.4 (5)	3.5 (7)
Unclassifiable drugs	7.4 (15)	7.8 (16)	10.2 (21)	5.6 (11)
Number of patients	204	204	206	198

*P value < 0.10 but ≥ 0.05 from Fisher exact test of drug-placebo comparison.

†P value < 0.05 from Fisher exact test of drug-placebo comparison.

TABLE 13

Per cent of patients for whom medication was prescribed for the indicated reason during one or more follow-up periods

Reason	(Numerators given in parentheses) PLBO	TOLB	ISTD	IVAR
Part A. Metabolic disorders				
Gout	0.5 (1)	1.5 (3)	1.5 (3)	1.5 (3)
Hyperthyroidism	0.0 (0)	0.5 (1)	0.5 (1)	1.0 (2)
Hypothyroidism	2.0 (4)	3.4 (7)	2.4 (5)	3.5 (7)
Unspecified thyroid maladies	0.0 (0)	2.5 (5)*	1.5 (3)	1.5 (3)
Other metabolic disorders	0.5 (1)	1.5 (3)	0.5 (1)	1.5 (3)
Part B. Infections and related problems				
Abscesses and inflammations	5.9 (12)	5.9 (12)	6.8 (14)	2.0 (4)*
Ear infections	2.0 (4)	3.9 (8)	2.9 (6)	2.0 (4)
Female genital infections	6.4 (13)	5.9 (12)	2.9 (6)	2.0 (4)*
Gastrointestinal infections	8.3 (17)	3.4 (7)*	6.8 (14)	5.1 (10)
Male genital infections	1.0 (2)	1.5 (3)	0.0 (0)	2.0 (4)
Urinary infections	18.6 (38)	17.2 (35)	14.6 (30)	17.7 (35)
Other infections	9.8 (20)	7.8 (16)	9.2 (19)	9.1 (18)
Part C. Respiratory ailments				
Asthma, hay fever, and allergies	7.4 (15)	6.9 (14)	7.3 (15)	6.6 (13)
Emphysema	1.5 (3)	0.5 (1)	1.5 (3)	0.0 (0)
Pulmonary disease	7.4 (15)	8.3 (17)	4.9 (10)	6.6 (13)
Other respiratory ailments	70.6 (144)	72.5 (148)	70.9 (146)	70.7 (140)
Part D. Pain				
Arthritis and rheumatism	23.0 (47)	28.4 (58)	29.1 (60)	30.3 (60)
General aches and pains	74.0 (151)	68.1 (139)	73.3 (151)	78.3 (155)
Postoperative pain	4.4 (9)	2.5 (5)	1.9 (4)	2.0 (4)
Part E. Problems related to the reproductive system				
Contraception	1.5 (3)	1.5 (3)	1.5 (3)	1.5 (3)
Menopause	3.9 (8)	5.4 (11)	5.3 (11)	6.1 (12)
Menstrual irregularities	3.4 (7)	1.5 (3)	1.9 (4)	2.0 (4)
Pregnancy and complications	1.0 (2)	1.0 (2)	1.5 (3)	1.5 (3)
Part F. Gastrointestinal ailments				
Gastrointestinal distress	38.7 (79)	42.6 (87)	39.3 (81)	44.4 (88)
Other gastrointestinal problems	2.5 (5)	4.4 (9)	3.4 (7)	2.5 (5)
Part G. Surgery				
Gastrointestinal surgery	1.0 (2)	1.5 (3)	0.5 (1)	1.5 (3)
Genitourinary surgery	1.5 (3)	1.0 (2)	0.0 (0)	0.5 (1)
Thyroidectomy	0.5 (1)	1.5 (3)	0.0 (0)	1.0 (2)
Other surgery	3.4 (7)	2.9 (6)	2.9 (6)	2.5 (5)
Part H. Other reasons				
Anemia	7.4 (15)	4.4 (9)	6.3 (13)	6.6 (13)
Appetite suppression	1.5 (3)	2.5 (5)	1.5 (3)	2.0 (4)
Dermatologic problems	10.8 (22)	8.8 (18)	10.2 (21)	9.1 (18)
Edema	14.2 (29)	14.7 (30)	18.9 (39)	13.1 (26)
Fluid retention	3.4 (7)	4.9 (10)	2.9 (6)	4.5 (9)
Neuropathies	2.9 (6)	2.5 (5)	1.5 (3)	4.0 (8)
Noninfectious renal difficulties	2.9 (6)	3.4 (7)	2.9 (6)	6.1 (12)
Ophthalmologic problems	5.4 (11)	3.9 (8)	3.4 (7)	4.0 (8)
Peripheral vascular disease	0.5 (1)	2.9 (6)	1.0 (2)	1.0 (2)
Psychiatric problems	37.7 (77)	40.2 (82)	38.3 (79)	41.9 (83)
Trauma and environmental exposure	7.4 (15)	3.4 (7)	2.9 (6)*	4.0 (8)
Reason uncertain	22.5 (46)	21.6 (44)	21.8 (45)	18.7 (37)
Number of patients	204	204	206	198

*P value < 0.10 but ≥ 0.05 from Fisher exact test of drug-placebo comparison.

forms. The percentage of patients who reported hospitalization for cancer is given on the second line of part A of table 14. The percentage of patients who reported hospitalization for cancer and/or who had died and the coded cause of death was cancer is given on the fifth line of part A of table 14. Fewer patients in the tolbutamide treatment group than in the placebo treatment group had cancer; the Fisher exact test of the tolbutamide-placebo comparison of the number of patients with cancer yielded a P value of 0.03. The P values for the ISTD versus PLBO and IVAR versus PLBO comparisons were 0.46 and 0.50, respec-

TABLE 14

Per cent of patients with cancer

Part A. Per cent of patients with cancer (number of patients with a specified event given in parentheses)

	PLBO	TOLB	ISTD	IVAR
Number of patients at risk	205	204	210	204
Specified event				
Hospitalization for cancer	3.4 (7)	1.0 (2)	2.4 (5)	3.4 (7)
Status 10/31/69				
Alive	1.5 (3)	0.5 (1)	1.9 (4)	2.9 (6)
Dead of cancer	1.5 (3)	0.5 (1)	0.5 (1)	0.5 (1)
Cancer death only	2.4 (5)	0.5 (1)	1.4 (3)	0.5 (1)
Any cancer	5.9 (12)	1.5 (3)	3.8 (8)	3.9 (8)
Case fatality in per cent	66.7	66.7	50.0	25.0

Part B. P values from Fisher exact tests comparing the observed distribution of patients with the specified event

Specified event	TOLB versus PLBO	ISTD versus PLBO	IVAR versus PLBO	TOLB versus ISTD	TOLB versus IVAR
Hospitalization for cancer	0.18	0.74	1.00	0.21	0.15
Cancer death only	0.22	0.70	0.22	0.93	0.49
Any cancer	0.03	0.46	0.50	0.16	0.06

tively. The P value for the TOLB versus IVAR comparison of any report of cancer was 0.06; more patients in the insulin variable-treated group than in the tolbutamide-treated group reported having cancer.

BLOOD GLUCOSE LEVELS

Analysis of the mean changes in fasting blood glucose levels for the cohort of patients followed through the 19th follow-up examination indicated—as might be expected—that variable doses of insulin were more effective in maintaining reduced blood glucose levels over the period covered by this analysis than any of the other hypoglycemic treatments studied. The fasting blood glucose values were obtained as part of the short GTT (details of the methods for this test are given in the paper on design, methods, and baseline results[1]). As may be seen from figure 8, there was a large drop in glucose levels in all treatment groups after the initiation of treatment. At the end of the first follow-up period, patients in the placebo-treated group showed a 12.9 per cent reduction in fasting glucose levels from the baseline mean in contrast to a 24.9 per cent reduction in the tolbutamide-treated group, a 20.7 per cent reduction in the insulin standard-treated group, and a 22.2 per cent reduction in the insulin variable-treated group. Patients in these cohorts showed a gradual return toward the baseline levels in all treatment groups, except for those in the insulin variable-treated group. The mean fasting blood glucose level recorded at the 19th follow-up examination for patients in the insulin variable treatment group was 19.5 per cent lower than at baseline. The corresponding values for the other treatment groups were 13.7 per cent above baseline for PLBO, 2.3 per cent above baseline for TOLB, and 4.3 per cent below baseline for ISTD.

The lower fasting blood glucose levels in the insulin variable-treated group were maintained during the course of follow-up by increasing the number of units of insulin prescribed. The mean number of units of insulin prescribed for patients in the insulin variable treatment group was 12 at the first quarterly follow-up examination and 35 at the 19th quarterly follow-up examination.

An analysis of blood glucose control was based on definitions of glucose control derived from those used by the members of the Joslin Clinic.[12] These definitions are outlined in table 15. The percentage of patients who had GOOD control was almost identical for

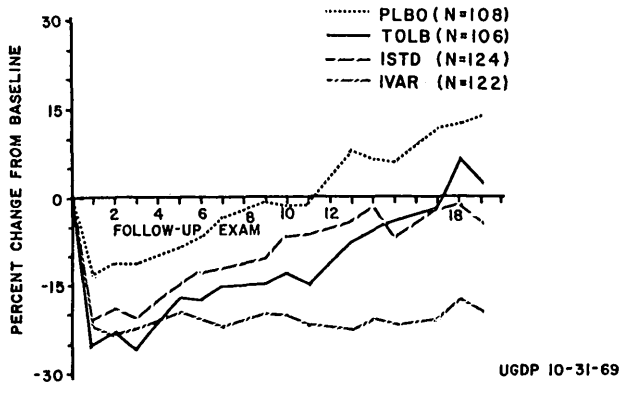


FIG. 8. Per cent change in fasting blood glucose levels from baseline to each follow-up examination for the cohort of patients followed through the 19th follow-up examination.

the tolbutamide- and insulin standard-treated groups, 37.4 for TOLB and 37.6 for ISTD. The highest levels of control were achieved in the insulin variable-treated group; 48.0 per cent were classified as having GOOD control and only 11.6 per cent were classified as having POOR control. The percentage of patients with POOR control in the other treatment groups was 41.2 for PLBO, 25.6 for TOLB, and 28.3 for ISTD.

TABLE 15

Percentage distribution of patients by level of blood glucose control

Level of control*	PLBO	TOLB	ISTD	IVAR
Good	27.0	37.4	37.6	48.0
Fair	31.9	36.9	34.1	40.4
Poor	41.2	25.6	28.3	11.6
Number of patients	204	203	205	198

*Definitions of blood glucose control are:

GOOD: 70 per cent or more of all the patient's short GTT fasting blood glucose values < 110 mg. per 100 ml.

POOR: 70 per cent or more of all the patient's short GTT fasting blood glucose values ≥ 130 mg. per 100 ml.

FAIR: Patient classified as having neither GOOD nor POOR blood glucose control.

ADHERENCE TO TREATMENT

The percentage of patients in each treatment group who took ALL of the assigned study medication for at least 75 per cent of all follow-up periods through October 31, 1969, is given in part A of table 16. For this calculation a patient was regarded as having taken ALL of the study medication in a given follow-up period if he/she was given a prescription for the assigned study medication in the dosage specified by protocol and if he/she took that medication for at least 75 days of that three-month period and did not take any hypoglycemic agent other than the assigned study medication during that period. This assessment of level of adherence was based on the patient's prescription record and the physician's judgment of the patient's adherence to that prescription. Nearly three-fourths (74 per cent) of the patients in the tolbutamide treatment group took ALL of their study medication for at least 75 per cent of the follow-up periods prior to the quarterly follow-up examination at which the prescription for tolbutamide was discontinued. The corresponding percentage for the placebo treatment group was 68.3 per cent. The lowest levels of adherence were noted in the insulin-treated groups (58.6 per cent for ISTD and 47.5 per cent for IVAR).

The per cent of patients who took ALL or SOME of the prescribed study medication for at least 75 per cent of all follow-up periods prior to the quarterly examination at which the protocol was changed is given in part B of table 16. For this analysis a patient was regarded as having taken SOME of the assigned medication if the patient was given a prescription for the assigned medication but for a dosage different from that specified by protocol (below or above that specified by protocol) and took this dosage for at least 75 days of the 90 days in that follow-up period or had a prescription for the exact dose specified by protocol and took this dose for at least 15 days but less than 75 days and if the patient had not taken any other hypoglycemic medication during this period. If the assessment of level of adherence is based on the per cent of follow-up periods in which patients took ALL or SOME of the prescribed medication, the results for the two insulin treatment groups are very similar to the results for the placebo and tolbutamide treatment groups.

The per cent of patients who took little or none of the assigned study medication during the entire time they were under study may be obtained from the last line of part B of table 16—that is, the percentage of patients who took ALL or SOME of the assigned study medication for less than 25 per cent of all follow-up periods. The per cent of patients who were classified

TABLE 16

Percentage distribution of patients by adherence to prescribed study medication

	Per cent of follow-up periods	PLBO	TOLB	ISTD	IVAR
Part A.	Per cent of patients who took ALL* of prescribed medication during a specified per cent of follow-up periods				
	≥ 75	68.3	74.0	58.6	47.5
	25–74	21.0	15.7	28.6	39.7
	< 25	10.7	10.3	12.9	12.7
Part B.	Per cent of patients who took ALL* or SOME† of prescribed study medication during a specified per cent of follow-up periods				
	≥ 75	72.2	79.4	73.8	77.9
	25–74	18.5	12.3	15.2	11.8
	< 25	9.3	8.3	11.0	10.3
Number of patients		205	204	210	204

*ALL: Patient took prescribed study medication in the dose specified by the protocol for at least 75 days during the follow-up period and took no other hypoglycemic agent.

†SOME: Patient took exact dose of prescribed study medication for at least 15 days or took a modified dose of the prescribed study medication for at least 75 days during the follow-up period and took no other hypoglycemic agent during the entire follow-up period.

in this way was 9.3 for PLBO, 8.3 for TOLB, 11.0 for ISTD, and 10.3 for IVAR.

LITERATURE REVIEW

A summary of animal studies, autopsy studies, and three prospective studies determining the long-term efficacy of tolbutamide was included in a previous UGDP report.[2] Since that report was published, Lasseter and co-workers[13] have shown an inotropic effect in human heart muscle on cardiac contractions and an increase in cardiac irritability due to tolbutamide and other sulfonylurea drugs. Wales, Grant, and Wolff[14] reported observing increased blood pressure levels in dogs following tolbutamide administration.

A report from the Kaiser-Permanente Drug Monitoring System indicated that there were more cases of congestive heart failure among tolbutamide users than among nonusers.[15] This difference was not statistically significant.

In a six-year prospective study of 186 newly diagnosed adult-onset diabetics by Boyle and co-workers,[16] the 71 patients treated predominantly with oral antidiabetic agents had a greater frequency of myocardial infarctions than the 115 patients treated predominantly with diet. These findings should be interpreted with some caution since patients were not randomly allocated to treatment but rather treatment was determined by the patient's blood sugar response.

The mortality experience of 184 patients with acute myocardial infarcts, confirmed by ECG and serum enzyme changes, and a known history of diabetes was reported recently by Soler et al.[17] At the time of admission to a coronary care unit, 27 of these 184 patients were being treated with dietary restrictions only and had never had any antidiabetic treatment other than diet. Ninety of the 184 patients were taking oral hypoglycemic agents (chlorpropamide, tolbutamide, glibenclamide, metformin, or a combination of a sulfonylurea and metformin), and 67 patients were taking insulin at the time of admission to the hospital. Patients on diet alone had an observed mortality of 15 per cent in the hospital, but in the groups receiving prescriptions for oral agents or insulin the observed mortality in the hospital was 40 per cent and 37 per cent, respectively. The mortality was slightly higher in the group treated with oral drugs even though the patients in the insulin group had a higher incidence of arterial hypertension, angina, and heart failure as well as retinopathy on admission with an MI and had been receiving antidiabetic treatment for a longer period of time. Mean duration of treatment with insulin was 12.9 years. The mean duration of treatment with oral drugs was 5.7 years, preceded by an average of 1.7 years of treatment with diet alone. These investigators also reported that 12 per cent of the patients receiving oral hypoglycemic drugs had primary ventrical fibrillation, as against 3 per cent of patients receiving insulin and 7 per cent of patients on diet alone. Although the number of patients in each of the groups was small and the lack of random assignment to treatment poses problems in the interpretation of these findings, these findings are consistent with observations made by Lasseter and Palmer[13] that showed an increase in cardiac irritability if a patient is taking one of the sulfonylureas and with the observation in the UGDP that there was an increased likelihood of a fatal outcome if the patient has a myocardial infarction and is taking tolbutamide than if the patient suffers this event and is taking insulin or no hypoglycemic agent.

Of the three prospective studies[18-20] reviewed in the earlier UGDP report,[2] only the study conducted by Keen and co-workers [18,21] has reported results on nonfatal vascular events. The results from this study will be compared with the findings from the UGDP in the next section.

The most recent report of the study of tolbutamide and phenformin conducted by Feldman and co-workers[19] noted that after 5.5 years of observation two deaths had occurred among 350 patients (174 assigned to tolbutamide treatment, 91 to phenformin treatment, and 85 to placebo treatment). No information on nonfatal events was presented.

Results from the Paasikivi study[20] of postcoronary patients indicated a lower mortality for patients treated with tolbutamide than for the placebo-treated group only during the first six months of follow-up. In the succeeding four years of follow-up the observed mortality in the tolbutamide-treated group was greater than that in the placebo-treated group. One reviewer concluded that "the report by J. Paasikivi in postcoronary patients is contradicting in its data and can more properly be interpreted as supporting the findings of the UGDP."[22]

COMPARISON OF RESULTS FROM BEDFORD STUDY AND RESULTS FROM UGDP

In all that has been written to date concerning the use of sulfonylurea drugs there is no evidence of beneficial effects associated with long-term use of tol-

butamide. Although the early report by Keen and co-workers[18] on the Bedford study has been cited as providing evidence of beneficial results from tolbutamide, this was not borne out by later analyses by these investigators.[21] Patients in this study were randomly assigned to placebo or to tolbutamide therapy. The 123 patients assigned to tolbutamide therapy received a fixed dosage of 1.0 gm. of tolbutamide per day (0.5 gm. twice daily).‡ The remaining 125 patients were treated with "placebo" tablets (two per day) each containing 3 mg. of tolbutamide. Each of these subgroups was further subdivided so that half were advised to reduce carbohydrate intake to 120 gm. daily and the other half only to reduce their intake of sugar. Every six months, patients returned for examination and a variety of standard measurements were made. Analysis of cardiovascular events observed during the first five years of follow-up indicated that the group treated with tolbutamide and advised to restrict carbohydrate intake had fewer cardiovascular events than did the patients in any of the other three groups. The number of events in the individual groups was small, but the percentage of patients with probable events in the two tolbutamide-treated groups (22 per cent) was lower (P value less than 0.05 without adjustment for multiple testing§) than observed in the two placebo-treated groups (33 per cent).

At the end of seven years of follow-up the proportion of patients with one or more cardiovascular events was lower in the tolbutamide treatment group than in the placebo treatment group, but Keen and coworkers did not consider this difference "significantly" lower (P value < 0.10 but ≥ 0.05 without adjustment§).[21] For both these analyses, the investigators included the following events: probable myocardial infarction, probable angina pectoris, probable intermittent claudication, ECG worsening based on the Minnesota Code, stroke, and cardiovascular death.

In all previous UGDP reports the assessment of treatment effects has been based primarily on differences in the occurrence of single events since the interpretation of differences in treatment effects based on pooling events ranging in severity from intermittent claudication to death is not perspicuous. However, to provide the reader with results for comparison with the Bedford study, the findings for the same cardiovascular events (although there may be some minor differences in definitions) are presented for UGDP patients in part II of table 17; the reported results from the Bedford study are given in part I of the same table. The average length of follow-up in the UGDP, as of October 31, 1969, was approximately five years. Some patients had completed eight years and some less than four years, but 654 of the 823 patients studied had completed five years. The average length of follow-up in the Bedford study was seven years for the results presented in part I of table 17.

In the Bedford study the percentage of patients with each of the nonfatal cardiovascular events (based on all patients allocated to treatment, since no information is presented by Keen on the presence of these events at entry into the study) was higher in the placebo-treated group than in the tolbutamide-treated group except for myocardial infarctions and intermittent claudication. The percentage of patients with any fatal or nonfatal cardiovascular event was also higher for placebo-treated patients than for tolbutamide-treated patients, as noted previously. The findings for patients in the UGDP showed quite different trends. The percentage of patients with each of the nonfatal events (based on patients who did not have the event at baseline) was higher for patients in the tolbutamide-treated group than for patients in the placebo-treated group except for the occurrence of nonfatal myocardial infarctions and ECG worsening. The percentage of patients with any fatal or nonfatal cardiovascular event in the UGDP was 26.8 per cent for PLBO and 35.3 per cent for TOLB, in contrast to the Bedford study findings (37.4 for PLBO and 28.0 for TOLB).

It is unlikely that the results of two prospective studies would be identical since there are differences in design, in the patients studied, in definitions of endpoints, and in the hypoglycemic effects achieved by the specified treatments. Patients enrolled in the Bedford study were classified as "borderline" diabetics and would generally be considered to have less glucose intolerance or a milder disease than the patients studied in the UGDP. The patients in the Bedford study were about 10 years younger at the time of entry into the study than the UGDP patients, yet a larger proportion of patients in the placebo treatment group than in the tolbutamide treatment group in the Bedford study were over age 70 (29.6 per cent in PLBO,

‡In the first report from this study the number of patients assigned to tolbutamide was given as 122 and the number of patients assigned to placebo was 126.

§Any adjustment for multiple testing would increase the P value, making the results less "significant."

TABLE 17

Occurrence of cardiovascular events

I. Bedford Study Results, Keen and Jarrett[21]

Part A. Per cent of patients who experienced a specified cardiovascular event (Number of patients with a specified event given in parentheses)

Specified Event	PLBO	TOLB
Cardiovascular death	10.6 (13)	6.4 (8)
Myocardial infarction	4.1 (5)	5.6 (7)
Angina pectoris	12.2 (15)	6.4 (8)
Intermittent claudication	3.3 (4)	8.0 (10)
Stroke	3.3 (4)	1.6 (2)
ECG worsening	10.6 (13)	5.6 (7)
Any cardiovascular event listed above	37.4 (46)	28.0 (35)
Number of patients	123	125

Part B. P values from Fisher exact tests comparing the observed distribution of patients with the specified event

Specified event	TOLB versus PLBO
Cardiovascular death	0.34
Myocardial infarction	0.79
Angina pectoris	0.17
Intermittent claudication	0.18
Stroke	0.65
ECG worsening	0.23
Any cardiovascular event listed above	0.15

II. University Group Diabetes Program, as of October 31, 1969

Part A. Per cent of patients who experienced a specified cardiovascular event (Number of patients with a specified event given in parentheses)

Specified event	PLBO	TOLB	ISTD	IVAR
Cardiovascular death	4.9 (10)	12.7 (26)	7.1 (15)	7.8 (16)
Nonfatal myocardial infarction	3.9 (8)	2.0 (4)	1.5 (3)	0.5 (1)
Angina pectoris*	13.9 (27)	16.3 (31)	11.1 (21)	13.1 (25)
Intermittent claudication	8.8 (16)	15.4 (28)	11.1 (21)	10.0 (18)
Stroke	0.0 (0)	0.0 (0)	0.5 (1)	0.0 (0)
ECG worsening	6.6 (13)	6.2 (12)	5.9 (12)	7.8 (15)
Any cardiovascular event listed above	26.8 (55)	35.3 (72)	27.6 (58)	28.9 (59)

Part B. Denominators

Specified event	PLBO	TOLB	ISTD	IVAR
Cardiovascular death	205	204	210	204
Nonfatal myocardial infarction	204	204	206	198
Angina pectoris*	194	190	190	191
Intermittent claudication	182	182	190	180
Stroke	204	204	206	198
ECG worsening	196	193	203	193
Any cardiovascular event listed above	205	204	210	204

Part C. P values from Fisher exact tests comparing the observed distribution of patients with the specified event

Specified event	TOLB versus PLBO	ISTD versus PLBO	IVAR versus PLBO	TOLB versus ISTD	TOLB versus IVAR
Cardiovascular death	0.01	0.45	0.30	0.08	0.14
Nonfatal myocardial infarction	0.38	0.21	0.04	0.99	0.39
Angina pectoris*	0.61	0.49	0.93	0.18	0.46
Intermittent claudication	0.08	0.58	0.83	0.28	0.17
Stroke	1.00	1.00	1.00	1.00	1.00
ECG worsening	1.00	0.93	0.81	1.00	0.69
Any cardiovascular event listed above	0.08	0.94	0.72	0.12	0.20

*History of angina pectoris from annual follow-up heart examinations and/or use of antianginal drugs or reason for drug prescribed reported as angina pectoris at quarterly follow-up examinations.

17.9 per cent in TOLB, P value < 0.05). Fixed doses of tolbutamide were used in both studies, but in the UGDP 1.5 gm. per day (the level most commonly used in clinical practice[1]) and in the Keen study 1.0 gm. per day was prescribed. Analysis of the results for capillary blood sugar levels two hours after a 50-gm. challenge indicated no differences among treatment groups in the Bedford study during the course of follow-up; this indicated no effect at all on two-hour postchallenge blood sugar of 1.0 gm. of tolbutamide, while a marked hypoglycemic effect was obtained with the dose used in the UGDP. Whether these

differences or random variation account for the conflicting results from these two studies is unclear.

Perhaps it should be noted that the placebo group in the Bedford study took a small amount of tolbutamide. Although there is little likelihood that this had an effect on blood glucose, it is not clear what effect the presence of a potentially sensitizing or allergenic agent might have. Bloodworth et al.[23] found microgranulomata in heart muscle of patients treated with sulfonylurea agents. These findings have not been confirmed by others;[24,25] thus, it seems unlikely that the detrimental effects of tolbutamide are due to sensitization. However, if this was the case, then no differences would be expected in the groups studied by Keen, since both the treated group and the control group received some of the agent.

The Committee for the Assessment of Biometric Aspects of Controlled Clinical Trials of Hypoglycemic Agents[26] concluded that the only mortality data extensive enough to determine the efficacy of these agents were those from the UGDP study and the Bedford study. This committee noted that many of the criticisms leveled against the UGDP would apply "sometimes even more strongly, to the Bedford study, but it was the UGDP trial that was challenged by the critics." After reviewing the published evidence available at the end of 1972 and reanalyzing some of the data from the UGDP and Bedford studies, the Biometrics Committee concluded "we consider that, in the light of the UGDP findings, it remains with the proponents of the oral [agents] to conduct scientifically adequate studies to justify the continued use of such agents."

DISCUSSION AND SUMMARY

In this report, a review of the development of nonfatal events has been presented for patients assigned to placebo, tolbutamide, insulin standard, or insulin variable based on observations reported at all quarterly examinations completed up to and including the quarterly examinations at which tolbutamide therapy was discontinued for tolbutamide patients and the examinations completed in the same interval for patients in the other treatment groups. The emphasis in the earlier UGDP reports on tolbutamide therapy was on the analysis of mortality data because results for nonfatal events are not so important if significant differences among the treatment groups have been observed for mortality. Review of all the findings for fatal and nonfatal events provides no evidence of benefit associated with the long-term use of tolbutamide.

The comparability of the treatment groups was established in earlier reports. More important, the observed tolbutamide-placebo differences in mortality and cardiovascular mortality changed very little when the baseline characteristics of the patients in these groups were taken into account in the analysis.[2,26,27] Of course, it was not possible to adjust for risk factors not measured, such as smoking history.// However, the effect of an extreme difference in the composition of the placebo and tolbutamide groups with respect to smoking history, such as 30 per cent in PLBO and 50 per cent in TOLB, which has only one chance in 50,000 of occurring if 40 per cent of all UGDP patients were smokers, could account for only 1.6 cardiovascular deaths or only 16 per cent of the observed excess cardiovascular mortality.[27]

Except for blood glucose levels, few differences in the occurrence of nonfatal events or conditions determined at scheduled follow-up examinations were observed among the groups treated with diet alone, tolbutamide and diet, or insulin and diet in the UGDP. The fasting blood glucose levels in the tolbutamide-treated group were lower than the levels in the diet alone group and about the same as the levels in the group treated with a fixed dose of insulin; patients treated with variable doses of insulin had consistently lower levels than the patients in the other three groups (figure 8 and table 15.). None of the treatments had much effect on body weight, although the patients in the group treated with diet alone lost a little more weight than the patients in the other groups (figure 6). In contrast to the findings for body weight, a small decrease in serum cholesterol levels was observed in the insulin variable treatment group and a small increase in the placebo treatment group after one year of follow-up (figure 7). The serum cholesterol levels in the insulin variable-treated group remained below the baseline levels for almost five years of follow-up. During this same period further increases were observed in the placebo-treated group. The possibility of long-term prevention of coronary heart disease as a result of the effect of insulin on cholesterol levels in the insulin variable-treated group cannot be dismissed, but the observed cardiovascular mortality in these two treatment groups during the period of follow-up covered

//Some information on smoking history was collected retrospectively in the UGDP. These data will be presented in the UGDP report on risk factors, which is now in preparation.

by this analysis provided no significant evidence of a beneficial effect of variable doses of insulin.

Regardless of the definition used for hypertension, the trends were about the same in all treatment groups, indicating no effect of blood glucose changes on blood pressure levels. Fewer patients in the placebo-treated group than in any of the other groups reported taking digitalis or medication for congestive heart failure (table 6). An increased risk of congestive heart failure among diabetics compared with nondiabetics was observed in the Framingham study.[28] Extensive analysis of the Framingham data indicated that the excessive risk was confined to subjects treated with insulin; no discernible risk was observed for subjects treated with tolbutamide or diet. In the UGDP the reported use of digitalis or medication for congestive heart failure by patients in the tolbutamide treatment group was almost the same as that for patients in the insulin standard treatment group but slightly more than that for patients in the insulin variable treatment group.

The occurrence of myocardial infarctions as well as the other vascular changes studied in this report was about the same in all treatment groups even though differences in blood glucose levels were observed. These results suggest that attempts to normalize blood glucose levels in the adult-onset diabetic will not alter the incidence of renal impairment, retinal changes, or the other common complications of diabetes. Of course, it must be recognized that few patients were found to have renal or ocular disorders over the period covered by this report, and longer periods of follow-up may yield different results with respect to these vascular changes.

Fewer patients in the tolbutamide treatment group than in the other three groups were hospitalized for cancer or died and cancer was considered the primary cause of death (table 14). The Fisher exact test of the TOLB versus PLBO and TOLB versus IVAR comparisons of the occurrence of cancer yielded P values < 0.10. These findings suggest the principle of competing risks. Thus, if patients treated with tolbutamide had more severe myocardial infarctions, it may be assumed that they did not have a chance to develop cancer.

The excess mortality observed in the tolbutamide-treated group over that of patients treated with diet or diet plus insulin appeared to be a result of increased mortality among tolbutamide-treated patients who had sustained a myocardial infarction (table 5). The mortality for all patients who had at least one myocardial infarction during the course of follow-up was 50 per cent for patients in the tolbutamide-treated group and 18 per cent for patients in the placebo-treated group. The mortality for such patients in the insulin-treated groups was 35 per cent for ISTD and 40 per cent for IVAR. These observations are consistent with the observations that sulfonylurea drugs may produce an enhanced inotropic effect on heart muscle and increased automaticity of the Purkinje fibers, since these factors may increase the extent of ischemic myocardial damage and lead to arrhythmias.[13] The already mentioned report[17] from Great Britain of a greater occurrence of primary ventricular fibrillation among patients receiving prescriptions for sulfonylurea and/or biguanide drugs than for patients receiving prescriptions for insulin on admission to a coronary care unit because of a myocardial infarction add further support to this explanation of the observed excess cardiovascular mortality in the tolbutamide-treated group.

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UNIVERSITY GROUP DIABETES PROGRAM

Past and present participating investigators as of January 1976 were as follows:

Aguilo, Francisco, M.D., San Juan, Puerto Rico.
Albrink, Margaret J., M.D., Morgantown, West Virginia.
Barrett, James C., M.D., Birmingham, Alabama (until 1972).
Becker, Frank O., M.D., Chicago, Illinois.
Biern, Samuel, M.D., Williamson, West Virginia.
Boshell, Buris R., M.D., Birmingham, Alabama.
Bowen, Angela J., M.D., Seattle, Washington (until 1970).
Cammarn, Maxine R., M.D., Cleveland, Ohio (until 1965).
Crampton, Joseph H., M.D., Seattle, Washington (died 1966).
Daughaday, William H., M.D., St. Louis, Missouri.
Davidson, Paul C., M.D., Williamson, West Virginia.
Field, Richard A., M.D., Boston, Massachusetts (until 1966).
Goetz, Frederick C., M.D., Minneapolis, Minnesota.
Goldner, Martin G., M.D., Brooklyn, New York.
Haddock, Lillian, M.D., San Juan, Puerto Rico.
Jacobs, William H., M.D., Williamson, West Virginia (died 1971).
Jacobson, Maynard E., M.D., Minneapolis, Minnesota.
Jones, Charles A., M.D., Williamson, West Virginia (died 1974).
Kansal, P. C., M.D., Birmingham, Alabama.
Kilo, Charles, M.D., St. Louis, Missouri.
Klimt, Christian R., M.D., Dr.P.H., Baltimore, Maryland.
Knatterud, Genell L., Ph.D., Baltimore, Maryland.
Knowles, Harvey C., Jr., M.D., Cincinnati, Ohio.
Kreines, Kenneth, M.D., Cincinnati, Ohio.
Leon, Eloina, M.D., San Juan, Puerto Rico.
Levin, Marvin E., M.D., St. Louis, Missouri.

Mackenzie, Malcolm S., M.D., Cleveland, Ohio (until 1969).
Martin, Donald B., M.D., Boston, Massachusetts.
Maslansky, Robert A., M.D., Minneapolis, Minnesota (until 1973).
Meinert, Curtis L., Ph.D., Baltimore, Maryland.
Metz, Robert J., M.D., Seattle, Washington.
Miller, David I., M.D., Baltimore, Maryland.
Miller, Max, M.D., Cleveland, Ohio.
Newberry, William B., Jr., M.D., Cleveland, Ohio.
Nibbe, Albert F., M.D., Chicago, Illinois (1966-1969).
Nielsen, Robert L., M.D., Seattle, Washington.
Osborne, Robert K., M.D., Boston, Massachusetts.
Prout, Thaddeus E., M.D., Baltimore, Maryland.
Recant, Lillian, M.D., St. Louis, Missouri (until 1967).
Reeves, Robert L., M.D., Seattle, Washington (until 1970).
Rovira, Gabriel Martinez, M.D., San Juan, Puerto Rico (until 1971).
Schwartz, Theodore B., M.D., Chicago, Illinois.
Spergel, Gabriel, M.D., Brooklyn, New York.
Steenrod, William J., M.D., Seattle, Washington.
Tucker, Randolph, M.D., Chicago, Illinois (until 1966).
Vega, Luis A., M.D., San Juan, Puerto Rico (until 1970).
Villavicencio, Elena, M.D., San Juan, Puerto Rico (until 1967).
Weisenfeld, Shirley, M.D., Brooklyn, New York.

Participating consultants included the following:

Jacob E. Bearman, Ph.D., statistics, University of Minnesota, Minneapolis, Minnesota.
Henry Blackburn, M.D., electrocardiography, University of Minnesota, Minneapolis, Minnesota.
Byron W. Brown, Jr., Ph.D., statistics, Stanford University, Palo Alto, California.
Jerome Cornfield, statistics, George Washington University, Washington, D.C.
Alan Freemond, M.D., ophthalmology, University of Cincinnati, Cincinnati, Ohio (until 1968).
Philip M. LeCompte, M.D., pathology, Faulkner Hospital, Boston, Massachusetts.
Alexander Lewitan, M.D., radiology, Kingsbrook Jewish Medical Center, Brooklyn, New York.
Irving M. Liebow, M.D., electrocardiography, University Hospital of Cleveland, Cleveland, Ohio (until 1966)
J. Wallace McMeel, M.D., ophthalmology, Retina Associates, Boston, Massachusetts.
Frederick A. Rose, M.D., radiology, University Hospitals of Cleveland, Cleveland, Ohio.

APPENDIX A
ECG Codes

1. Definitions of Significant Worsening

Q/QS patterns

a. Occurrence of findings coded 1-1-1 to 1-2-7 on a follow-up ECG in patients with normal Q/QS on the baseline ECG;
b. Occurrence of findings coded 1-1-1 to 1-1-7 on a follow-up ECG in patients with findings coded 1-2-8 or 1-3-1 to 1-3-7 on the baseline ECG.

Q/QS and T patterns

One of the following changes for Q/QS:

a. ECG findings coded 1-1-1 to 1-3-7 on a follow-up ECG with normal Q/QS on the baseline ECG;
b. ECG findings coded 1-1-1 to 1-2-7 on a follow-up ECG in patients with findings coded 1-2-8 or 1-3-1 to 1-3-7 on the baseline ECG;
c. ECG findings coded 1-1-1 to 1-1-7 on a follow-up ECG in patients with findings coded 1-2-1 to 1-2-7 on the baseline ECG;

and one of the following changes in T patterns:

a. ECG findings coded 5-1 to 5-3 on a follow-up ECG in patients with normal T patterns on the baseline ECG;
b. ECG findings coded 5-1 or 5-2 on a follow-up ECG in patients with findings coded 5-3 on the baseline ECG;
c. ECG findings coded 5-1 on a follow-up ECG in patients with findings coded 5-2 on the baseline ECG.

2. Minnesota Code for Resting Electrocardiogram

Q and QS patterns

(Do not code in presence of ventricular conduction defects 6-4 or 7-1.)

1-1-1 Q/R amplitude ratio 1/3 or more plus Q duration 0.03 sec. or more in any of leads I, II, V2, 3, 4, 5, 6.
1-1-2 Q duration 0.04 sec. or more in any of leads I, II, V1, 2, 3, 4, 5, 6.
1-1-3 Q duration 0.04 sec. or more, plus R amplitude of 3 mm. or more in lead aVL.
1-1-4 Q duration 0.05 sec. or more in lead III plus any Q wave of at least 1.0 mm. amplitude in aVF.

1-1-5 Q duration 0.05 sec. or more in lead aVF.
1-1-6 QS pattern when R wave is present in adjacent lead to the right on the chest in any of leads V2, 3, 4, 5, 6.
1-1-7 QS pattern in all of leads V1-V4, V1-V5, or V1-V6.
1-2-1 Q/R amplitude ratio 1/3 or more, plus Q duration at least 0.02 sec. and less than 0.03 sec. in any of leads I, II, V2, 3, 4, 5, 6.
1-2-2 Q duration at least 0.03 sec. and less than 0.04 sec. in any of leads I, II, V2, 3, 4, 5, 6.
1-2-3 QS pattern in lead II.
1-2-4 Q duration of at least 0.04 sec. and less than 0.05 sec. in lead III, plus any Q wave of at least 1.0-mm. amplitude in aVF.
1-2-5 Q duration at least 0.04 sec. and less than 0.05 sec. in lead aVF.
1-2-6 Q amplitude of 5 mm. or more in either of leads III, aVF.
1-2-7 QS pattern in all of leads V1 through V3.
1-2-8 R amplitude decreasing to 2 mm. or less, and absence of codes 3-2, 7-2, or 7-3, between any of leads V2 and V3, V3 and V4, V4 and V5, or V5 and V6.
1-3-1 Q/R amplitude ratio at least 1/5 and less than 1/3 plus Q duration of at least 0.02 sec. and less than 0.03 sec. in any of leads I, II, V2, 3, 4, 5, 6.
1-3-2 QS pattern in absence of code 3-1, in each of leads V1 and V2.
1-3-3 Q duration of at least 0.03 sec. and less than 0.04 sec., plus R amplitude of 3 mm. or more in lead aVL.
1-3-4 Q duration of at least 0.03 sec. and less than 0.04 sec. in lead III, plus any Q wave of at least 1.0-mm. amplitude in lead aVF.
1-3-5 Q duration of at least 0.03 sec. and less than 0.04 sec. in lead aVF.
1-3-6 QS pattern in each of leads III and aVF.

T wave items

(Do not code in presence of ventricular conduction defects 6-4, 7-1, 7-2, 7-4.)

5-1 T amplitude negative, minus 5 mm. or more in any of leads I, II, V2, 3, 4, 5, 6 or in lead aVL when R amplitude is 5 mm. or more, or in lead aVF when QRS is mainly upright.
5-2 T amplitude negative or diphasic (positive-negative or negative-positive type) with negative phase at least minus 1.0 mm. but not as deep as minus 5 mm. in any of leads I, II, V2, 3, 4, 5, 6, or in lead aVL when R amplitude is 5 mm. or more, or in lead aVF when QRS is mainly upright.
5-3 T amplitude zero (flat), or negative, or diphasic (negative-positive type) with less than 1.0 mm. negative phase in any of leads I, II, V3, 4, 5, 6, or in lead aVL when R amplitude is 5 mm. or more; not coded in lead aVF.
5-4 (Optional code): T amplitude positive and T/R amplitude ratio less than 1/20 in any of leads I, II, aVL, V3, 4, 5, 6; R-wave amplitude must be 10 mm. or more.